Fiscal Year:	FY 2021	**Task Last Updated:**	FY 10/27/2020
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Project Title:	NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	01/01/2016	**End Date:**	12/31/2021
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2021 per NSSC information (Ed., 2/1/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AB52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL," grant number NNX10AU19G. Funding is for Dr. Kelton's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main project: Thermophysical and structural properties and solidification behavior of supercooled glass forming liquids as a member of THERMOLAB; the Science Coordinator is Prof. Dr. Hans Fecht, Ulm University, Institute of Micro and Nanomaterials, Albert-Einstein-Allee 47, Ulm Germany. Similar measurements on undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); the Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum fur Luft- und Raumfahrt (DLR), Institut fur Materialphysik im Weltraum, 51170 Köln, Germany. Subprojects: 1. Ground-Based Experimental Program: Synchrotron X-ray studies of liquid structures are made at the Advanced Photon Source using the ESL (electrostatic levitation) chamber at Washington University (WU-BESL), which has been optimized for scattering studies. These are correlated with neutron scattering data taken at the Spallation Neutron source using the Neutron ESL (NESL), with thermophysical property data obtained on ground with WU-BESL and using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) on the International Space Station (ISS). Terrestrial maximum undercooling and induction time to recalescence data are obtained in ESL, where stirring ensures that nucleation is not diffusion controlled. This data will allow an evaluation of diffusion in the upcoming MSL-EML ISS undercooling studies. 2. Thermophysical Property Measurements: ISS. Specific heat, density, surface tension, viscosity, and electrical resistivity measurements are made on supercooled glass-forming liquids and of Ti-Zr-Ni liquids with different Ni concentrations as a function of the temperature using the MSL-EML facility. These are correlated with x-ray synchrotron and elastic neutron diffraction studies of the supercooled liquids and ground-based measurements made using WU-BESL The purpose is to determine the existence and nature of a cooperative cross-over behavior in the shear viscosity at high temperature, which may be associated with chemical and topological ordering and a possible connection with liquid fragility. Measurements of the specific heat and electrical resistivity are also of interest for their correlation with the dynamical crossover. 3. Effect of diffusion on nucleation: The importance of long-range diffusion on nucleation is being evaluated using the MSL-EML facility located on the ISS. The critical items of interest are the maximum undercooling and the induction time to recalescence in isothermal holds at temperatures below the liquidus temperature. Together with the results from the thermophysical property measurements and an MHD convection model results this will allow us to analyze how the maximum undercooling is influenced by fluid flow. These results of the micro-g experiments are compared with the results of ground-based ESL measurements and with predictions from Kelton’s coupled-flux model for nucleation. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: 1. Ground Based Experiment Program. Measurements of the density, surface tension and viscosity, and synchrotron x-ray diffraction measurements of the liquid structure in the supercooled state using WU-BESL. Ground-based measurements of the maximum undercooling and induction time to recalescence in isothermal holds are also made using WU-BESL. These will be used to evaluate the nature of ordering on phase transitions in the supercooled liquid and fragility. 2. Long Duration Microgravity Experiments. Measurements of thermophysical properties, specific heat (modulated measurements), surface tension, viscosity and density, and also electrical resistivity are made as a function of temperature using the MSL-EML on the ISS. The results are coupled with x-ray and elastic neutron diffraction measurements of the liquid structure factor (WP 2) and ground based studies of the density and viscosity to study the role of ordering on novel liquid/liquid phase transitions and to determine the nature of these transitions. Maximum undercooling and time to recalesce measurements as a function of temperature and stirring will be compared with terrestrial ESL measurements to assess the importance of diffusion on nucleation, allowing Kelton's coupled flux model for nucleation to be evaluated experimentally.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which have the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science, encapsulated in the Materials Genome Initiative. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. Even containerless based methods, such as electrostatic levitation, cannot be used to accurately investigate these. The ISS experiments that are being carried out under the support of this grant are allowing these data to be obtained.

Task Progress:

Summary of activities during 2019-2020

1. Background. Over the past year our research has expanded from the two key questions discussed in previous reports to three. The first question is the investigation of the effects of stirring on crystal nucleation when the liquid and primary crystallizing phases have a different chemical composition. The second explores how the high temperature thermophysical properties of a metallic liquid may be relevant to glass formation. A third new question grew out of the analysis of measurements of the electrical resistivity at high temperatures, near and above the liquidus temperature. These data, which were discussed in the 2018-2019 annual report, show a plateau in the temperature dependence of resistivity at high temperature that is not predicted by any existing theory. It is important to determine whether this is a general feature of metallic liquids, or is specific to the two alloy liquids studied To address these questions, experimental measurements are being made on levitated metallic liquids using the microgravity environment of the International Space Station (ISS) and under terrestrial conditions using the technique of electrostatic levitation (ESL) facility in our laboratory at Washington University. A brief summary of the results obtained since the last annual report is provided in this report. The key accomplishments made over the past year are:

• The analysis of new experimental data for nucleation from ISS and ground-based studies.

• An analysis of volume and thermal expansion data for metallic glass-forming liquids.

• An evaluation of the nucleation and growth data obtained from batch 1 and batch 2 experiments on the ISS.

• Preparatory steps taken for future measurements of batch 3 and batch 4 samples on the ISS.

2. Nucleation experiments

ISS studies - Nucleation is the first step in the crystallization of a liquid; nucleation data are typically analyzed using the Classical Theory of Nucleation (CNT). Our computer modeling studies have demonstrated that the kinetic model of the CNT is inadequate for nucleation of a crystal phase from a liquid with a different chemical composition. The coupled flux model for nucleation developed by the Principal Investigator-PI (Kelton) predicts that in those cases the nucleation rate will decrease with decreasing stirring in the liquid. Testing this prediction has been and remains a key goal of the ISS and terrestrial investigations made under the support of this grant. In both theories, the nucleation rate is described by the product of a pre-term, A*, and an exponential function of the work of cluster formation, W. For the same value of W, A* should be smaller if coupled-flux nucleation is dominant over CNT.

Estimates of these parameters can be obtained from the maximum amount that the liquid can be cooled below the melting temperature (maximum undercooling) before crystal formation. By taking data from many undercooling experiments, we obtained sufficient statistics to make these estimates. Undercooling measurements were made for two batch 1.3 liquids, Vit106 (Zr57Cu15.4Ni12.6Al10Nb5) and Ti39.5Zr39.5Ni21 and one batch 2 liquid, Cu50Zr50, under conditions of either minimum or maximum stirring which was induced by the RF-coils on the electromagnetic levitation facility on the ISS. From the results we were able to make the first assessment of the effect of stirring on nucleation. A clear increase in A* with increased stirring was observed in the Vit160 and Cu50Zr50 liquids. No statistically significant influence of stirring was found for the Ti39.5Zr39.5Ni21 liquid. However, in this case the primary nucleating phase is a quasicrystal that has very similar structure and composition as that of the liquid. Long range diffusion is therefore not required. Both of these observations are in agreement with predictions from the coupled flux model.

Ground-based studies – In last year's report ground-based nucleation rate measurements for Ti45Zr45Ni10 and Zr80Pt20, which have been selected for batch 3 experiments on the ISS, and for Ti40Zr30Ni30, which is closely related to the batch 4 sample Ti40Zr27Ni33, were discussed. This year measurements were made using our ESL on the batch 4 sample, Zr50Pd50 and two related alloys, Pd55Zr45 and Zr75.5Pd24.5. For comparison, a parallel study was made for similar compositions of the binary Cu-Zr alloys that are relevant to the ISS experiments. To obtain reliable statistics, the maximum undercooling was measured several hundred times for each alloy composition, as was done for the ISS measurements (although the number of measurements in the ESL was much larger than was possible in the EML on the ISS). The studies support our past conclusions that the structure of the liquid plays an important role for crystal nucleation. In almost all cases the liquids of these alloys contain significant icosahedral order. When the structures of the primary crystallizing phases have similar short-range order, the nucleation barrier is small; when the structures are not similar the nucleation barrier is large. What was surprising is that the pre-factor appeared to correlate with the nucleation barrier. The reasons for this are unclear and future studies are planned to investigate this.

3. Growth Velocity Measurements

The crystal growth velocity of the icosahedral quasicrystal phase was measured in a Ti39.5Zr39.5Ni21 liquid using the EML facility aboard the ISS and the growth velocity of a crystal phase mixture was measured in a Zr64Ni36 liquid using our ESL facility. In both cases, the growth rate was determined from high-speed video data of the advancing growth front. The data indicate that the growth mechanisms are very different in these two liquids. The growth velocity increases monotonically with decreasing temperature in Ti39.5Zr39.5Ni21 but shows a maximum with decreasing temperature in Zr64Ni36¬. The growth velocity in Ti39.5Zr39.5Ni21 can be fit by a layer growth model that is governed by the nucleation of new attachment surfaces on the growing interface (2D growth), while the Zr64Ni36 data fit best to a screw dislocation growth mechanism. The growth velocity of a quasicrystal from the liquid has not been directly measured before, so these new data will provide important insight into how this novel phase nucleates and grows.

4. Other ISS-related activities. The experiences gained from the batch 1 and batch experiments on the ISS have led to the development of new strategies for more effective experiments in future runs on the ISS using the MSL-EML facility. Two examples are listed below.

i. In the batch 2.1 studies, the Ti39.5Zr39.5Ni21 was processed first under vacuum followed by processing under Ar and He gas atmospheres. It was observed that contamination of the sample occurred during the gas processing, which was detrimental to the further planned studies. It was not possible to make the number of undercooling measurements to obtain sufficient statistics for a quantitative statement on the effects of stirring on nucleation In future batch samples, all nucleation measurements will be completed under vacuum before exposing samples to Ar and He.

ii. Molecular dynamics (MD) studies have suggested that cooperative dynamics began at a temperature (TA) that is near the liquidus temperature. The coherence length of the cooperativity increases with decreasing temperature. Our measurements of the electrical resistivity in two liquid metal alloys made using the EML on the ISS gave structural evidence for TA, in agreement with MD predictions. Further they showed that the temperature dependence of the

resistivity went to zero above TA, something that has not been predicted in any existing theory of the electrical resistivity of liquid metals. The two alloy liquids studied were in a cup; our measurements suggest a possible coupling between the RF-fields of the EML and the metal cages that are frequently used to contain the samples. Based on this information our batch 3 and batch 4 samples will be contained in cups, not cages.

iii. Other ISS studies are:

• Participation in the remote Investigators' Working Group (IWG)-13 meeting, May 13-14, 2020.

• An additional measurement campaign (batch 2.4) in August 2020 and October 2020 to complete experiments on a batch 2 sample (Ti39.5Zr39.5Ni21). Since the samples had been contaminated by exposure to Ar and He, measurements were made in He under different conditions to examine the stirring effects on nucleation for heterogeneous nucleation. Nineteen nucleation cycles were measured under the most quiescent conditions possible and nineteen were made under the most turbulent conditions.

• Batch 3 - QUASI is involved in experiments on four batch 3 samples. These are Cu47Zr47Al6 (ID#41) and Zr80Pt20 (ID#70) in which QUASI is the PI. QUASI is a co-PI on Ti45Zr45Ni10 (ID#50) (ICOPROSOL is the PI), and Pt57.5Cu14.7Ni5.3P22.5 (ID#81) (THERMOLAB is the PI). These were originally planned to be integrated into the sample chamber during the Summer/Fall of 2020 and flown to the ISS on a Space X rocket during the fourth quarter of 2020. Unfortunately, it is not now clear when these operations will be completed because of the COVID-19 pandemic. The Zr80Pt20 and Pt57.5Cu14.7Ni5.3P22.5 experiments are scheduled for batch 3.1, with the Ti45Zr45Ni10 sample scheduled for batch 3.2 and the Cu47Zr47Al6 sample in batch 3.3. The parameters for batches 3.2 and 3.3 are expected to be developed in 2020, either remotely or in-person at the DLR, depending on the COVID-19 situation.

• Batch 4 – The Ti40Zr27Ni33 (QUASI – co-PI) sample was selected for batch 4 and the Zr50Pd50 (QUASI – PI) was selected as a first back-up sample. The ground-based nucleation, viscosity, and thermal expansion studies have been completed. Samples of these compositions were prepared by QUASI for the parabolic flight tests, which were completed in October 2020. Both samples were successfully processed and have passed validation for the ISS experiments. If batch 3 processing on the ISS is successfully completed in 2021 and 2022, batch 4 studies may start in 2023.

5. Other related activities: Our research group has accumulated a large amount of volume and thermal expansion data for glass forming liquids over the past two decades. This year these data were correlated with glass formability in forty-one different liquids. The data showed that the molar volumes of the liquids that are better glass-formers are closer to those of the equilibrium crystal phases. Also, the volumes of better glass formers are smaller than expected from the constituent elemental liquids and correlate with a larger heat of mixing. These observations provide strong experimental evidence that efficient packing in the liquid is necessary for better glass formation. A similar correlation is found between the volume expansion coefficient of the liquids and glass formation, with the difference between the liquids and crystal phases smaller in the better glass formers. There are two contributing factors to the expansion coefficient in a liquid, the anharmonicity of the interatomic potential and changes in the liquid structure with temperature. Assuming that the anharmonicity is similar in the liquid and crystal phases, our data suggest that the liquid structural changes with decreasing temperature are smaller in the better glass forming liquids, which is consistent with our earlier X-ray scattering studies.

Bibliography Type:	Description: (Last Updated: 05/20/2025)
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